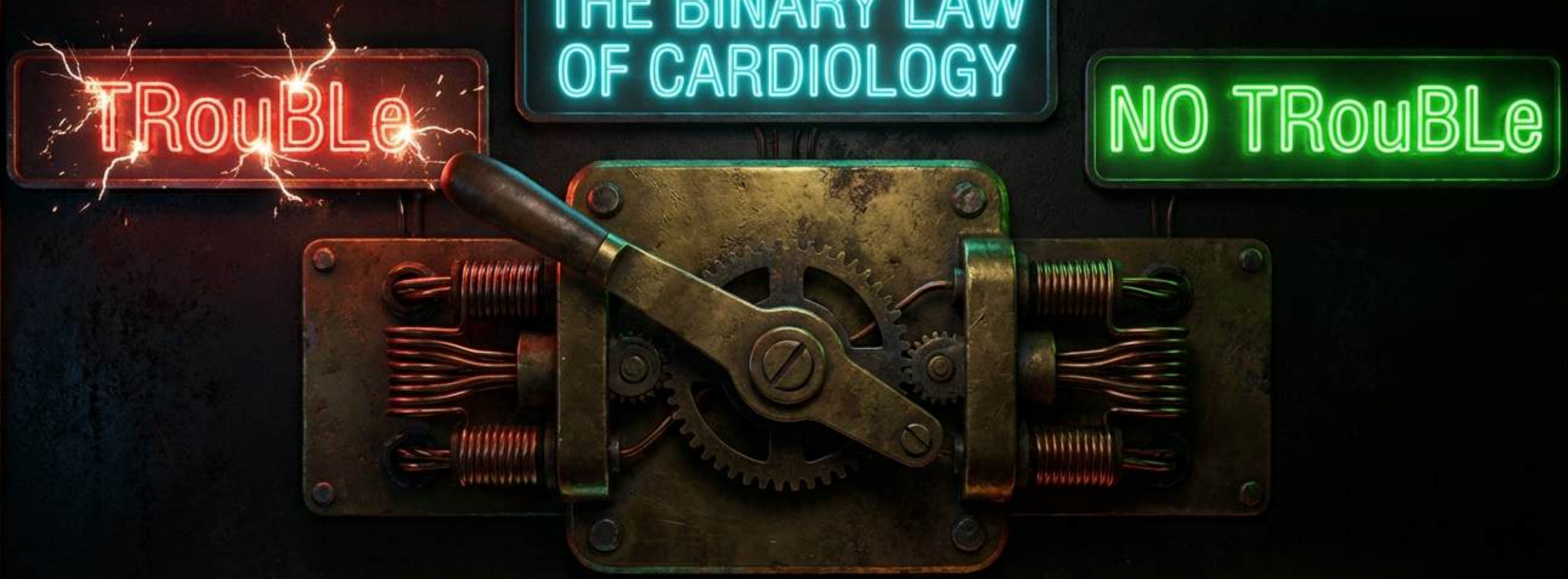
THE CONGENITAL SORTING ENGINE

A BINARY GUIDE TO PEDIATRIC HEART DEFECTS

OPERATIONAL MANUAL: THE 'TROuBLe' PROTOCOL

NotebookLM

THE BINARY LAW OF CARDIOLOGY

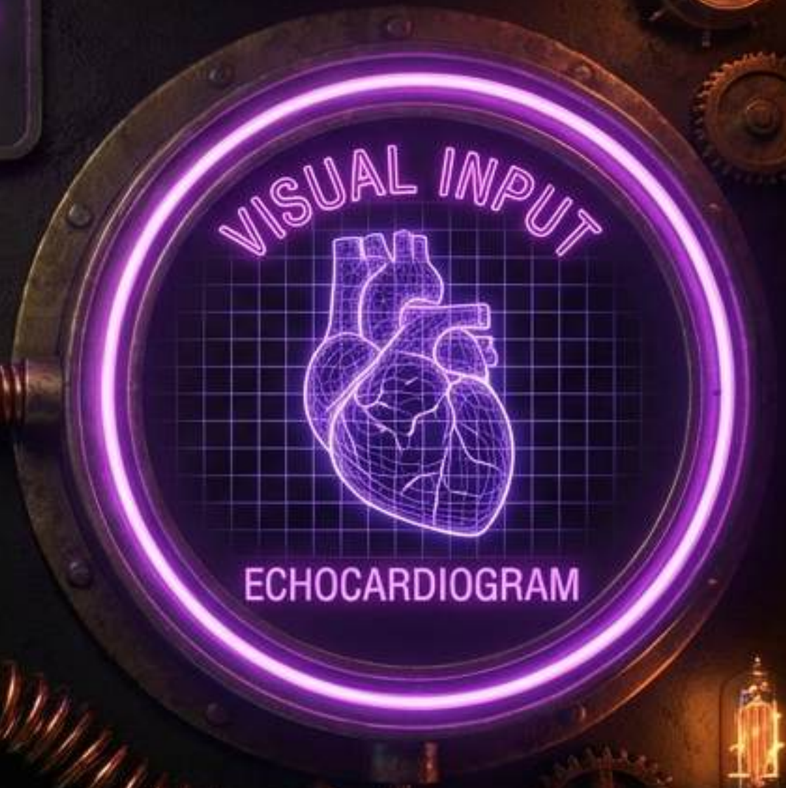


It either makes a lot of TRouBLe or No TRouBLe.

There is no in-between.

NotebookLM

THE UNIVERSAL CONSTANTS



Regardless of classification, every CHD child presents with:

1. A Murmur (due to blood shunting).
2. The need for an Echocardiogram (to find the cause).

NotebookLM

Rye

TRouLBe

Starts with the letter ‘T’

Right

Left

Blue

THE DIAGNOSTIC CODE

R-L-B are the capital letters.
Vowels are lowercase connectors.

NotebookLM

CRITICAL SYSTEM ERROR: R-to-L SHUNT

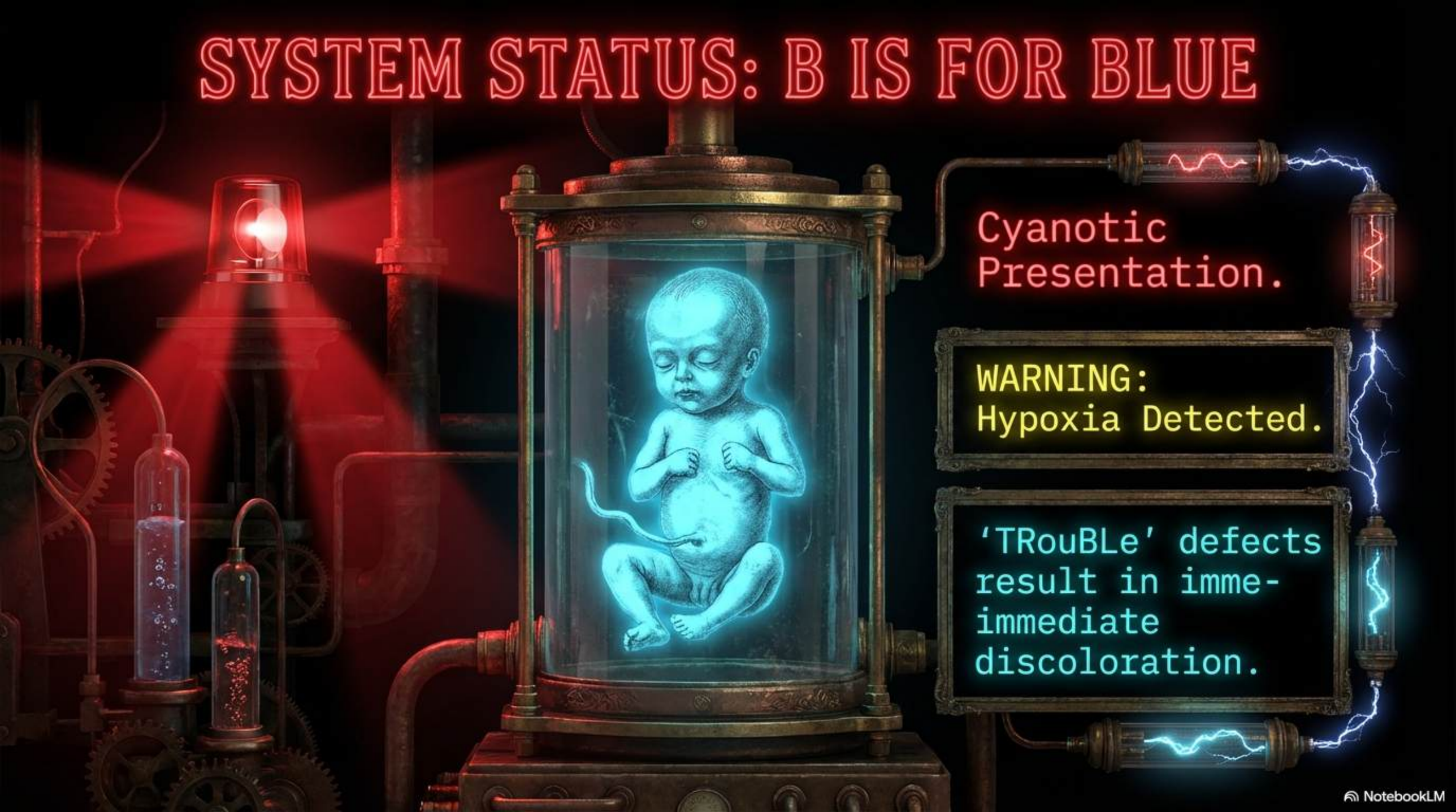
MECHANISM OF ACTION

Blood shunts Right to Left.

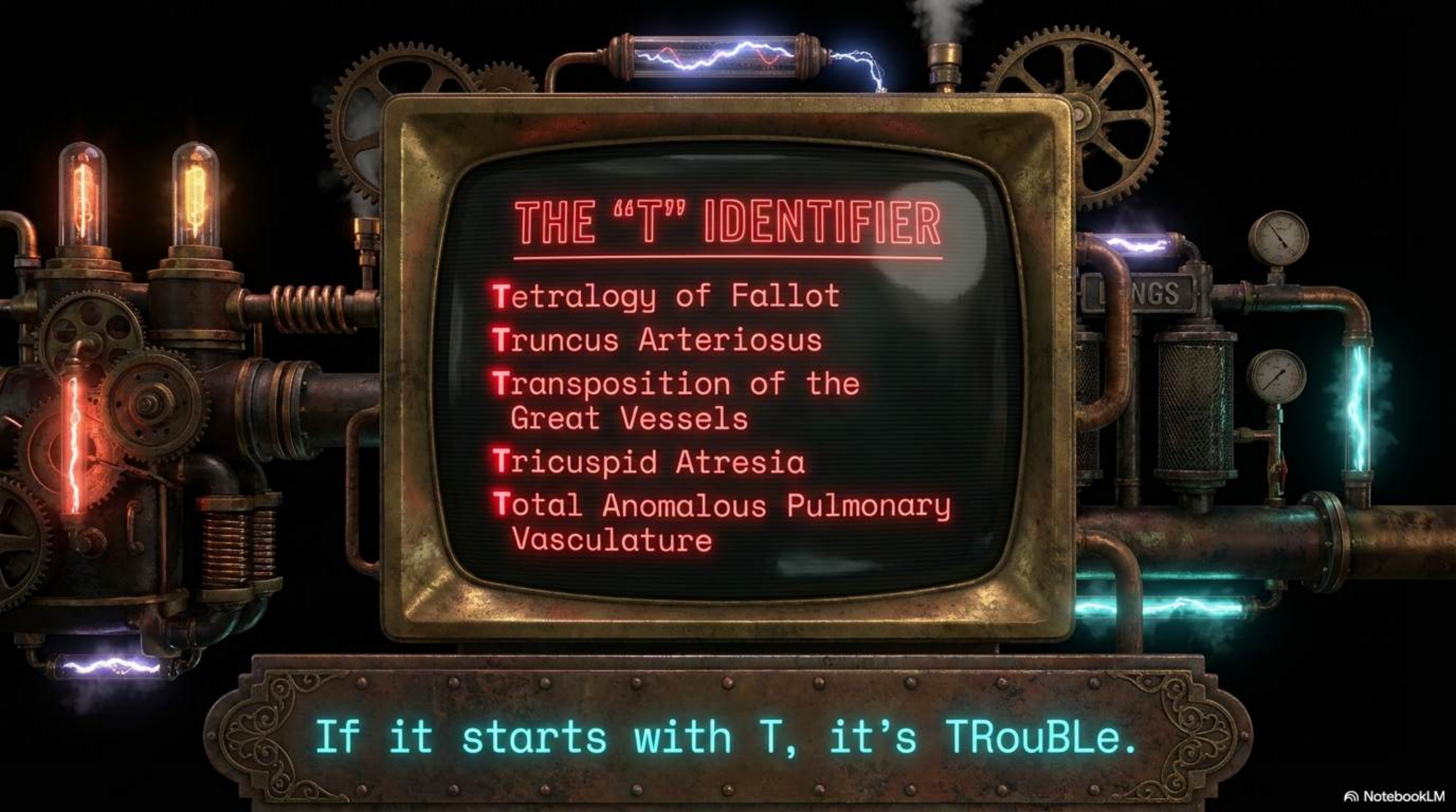
Unoxygenated blood enters systemic circulation.

NotebookLM

SYSTEM STATUS: B IS FOR BLUE



NotebookLM


THE "T" IDENTIFIER
Tetralogy of Fallot
Truncus Arteriosus
Transposition of the Great Vessels
Tricuspid Atresia
Total Anomalous Pulmonary Vasculature
If it starts with T, it's TRouBLe.

NotebookLM

EXCEPTION TO THE RULE

Left Ventricular Hypoplastic Syndrome

CAUTION CAUTION CAUTION

It starts with an 'L', but it acts like a 'T'.

This is a TRouBLe defect.

NotebookLM

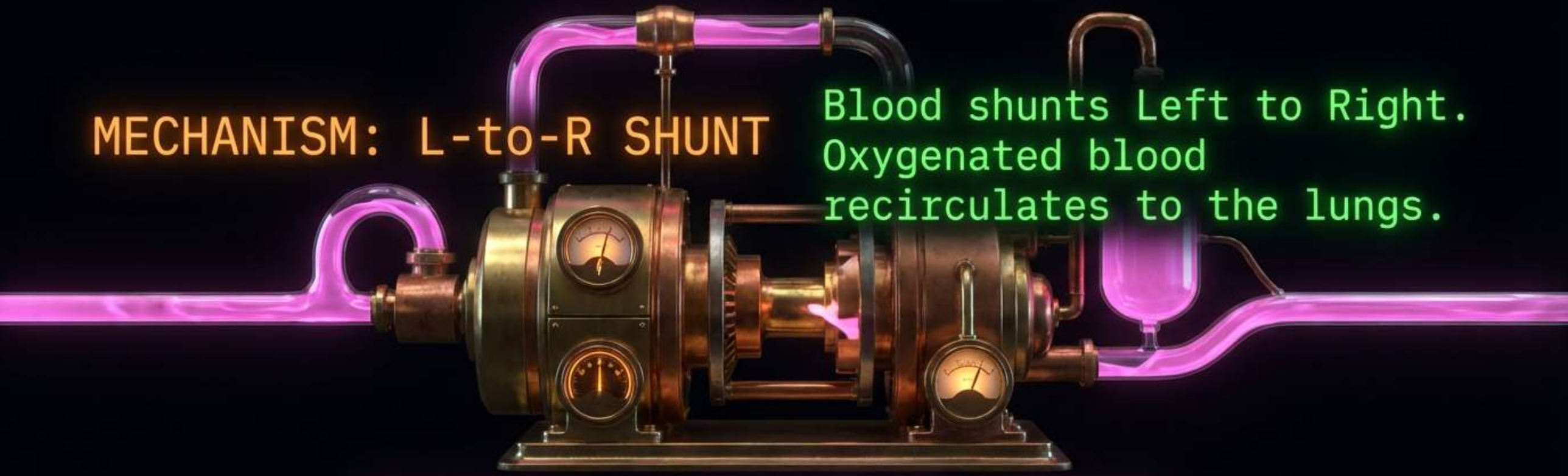
THE BURDEN OF "TRouBLe"

- SURGERY: Needed Immediately to live.
- GROWTH: Delayed / Failure to Thrive.
- LIFE EXPECTANCY: Shortened.
- FAMILY IMPACT: Grief, Financial Stress, Emotional Strain.
- DISCHARGE: Home on a Cardiac Monitor.

NotebookLM

PROTOCOL: NO TRouBLe

The “Safe” Category.



MECHANISM: L-to-R SHUNT

Blood shunts Left to Right. Oxygenated blood recirculates to the lungs.

Patient is ACYANOTIC (Pink).

NotebookLM

THE NON-'T' DEFECTS

- Ventricular Septal Defect (VSD)
- Patent Ductus Arteriosus (PDA)
- Patent Foramen Ovale
- Atrial Septal Defect
- Pulmonic Stenosis

If it doesn't start with T
(and isn't the L-exception), it's safe.

NotebookLM

TRouBLe

Shunt: R —> L
(Red Neon)

Color: Blue/Cyanotic
(Red Neon)

First Letter: Starts with T

Surgery: Immediate

Growth: Delayed

Home Equipment: Apnea Monitor

NO TRouBLe

Shunt: L —> R
(Green Neon)

Color: Pink/Acyanotic
(Green Neon)

First Letter: All Others

Surgery: None/Later

Growth: Normal

Home Equipment: None

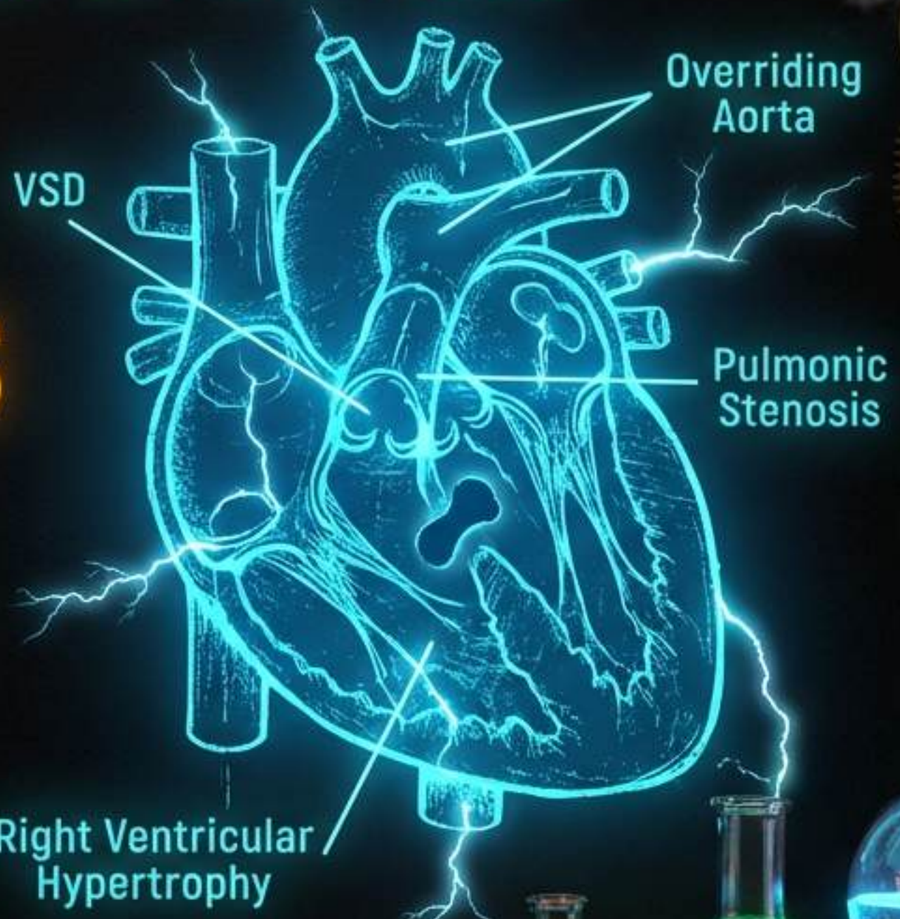
NotebookLM

TETRALOGY OF FALLOT

The 4 Defects hidden in the name

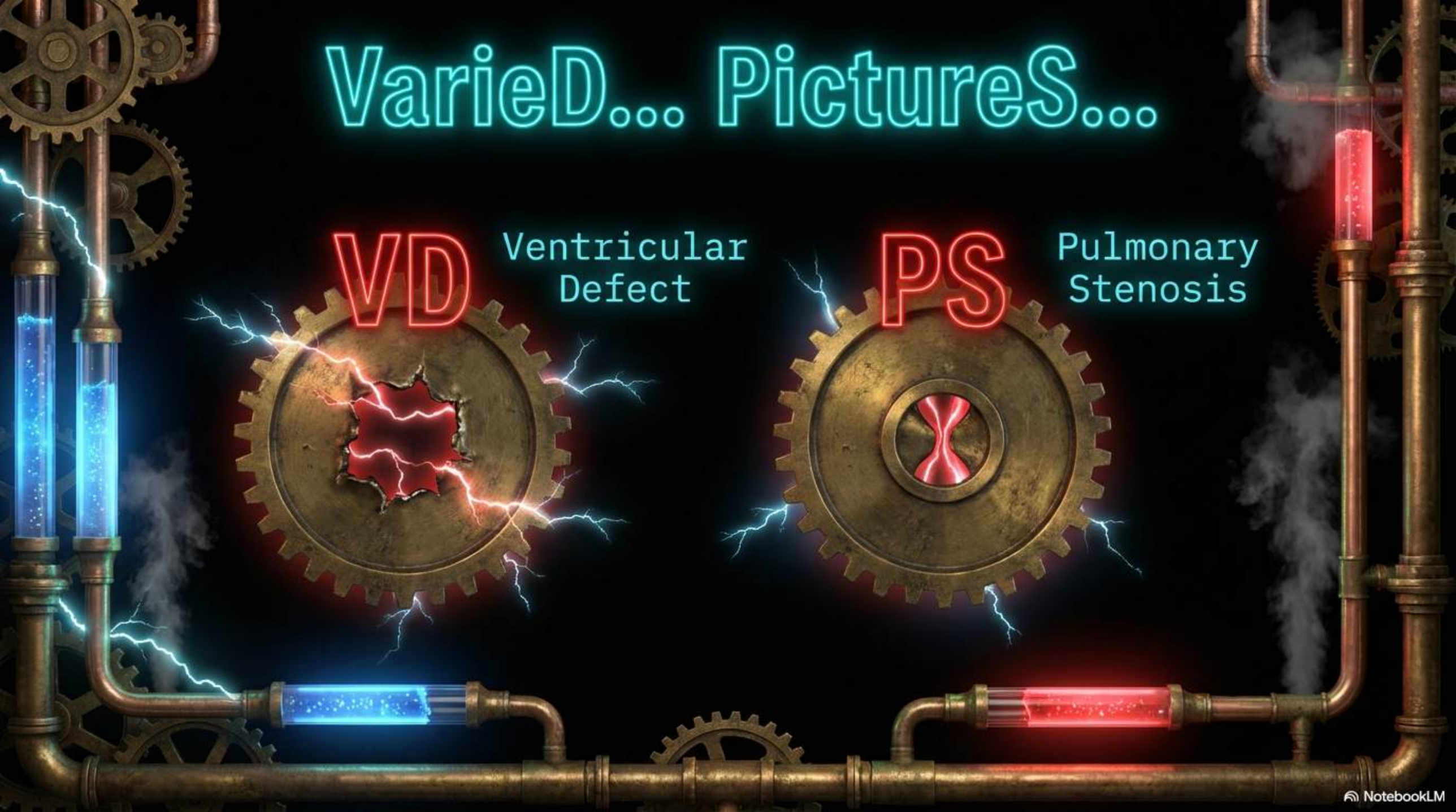
MNEMONIC: "VarieD PictureS Of A RancH"

Only the capitalized first letters matter



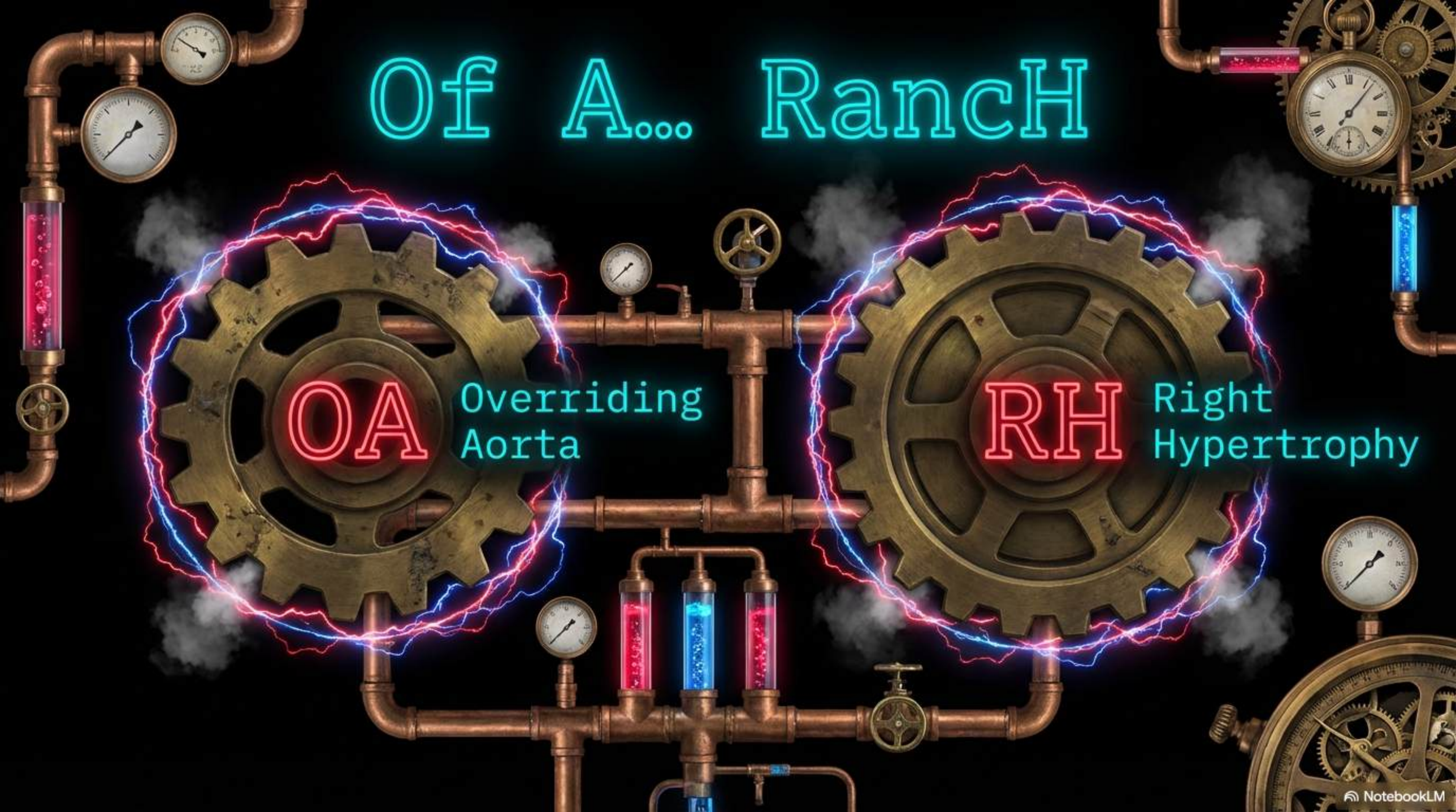
NotebookLM

VarieD... PictureS...



NotebookLM

Of A... RancH



NotebookLM

REMEMBER THE INITIALS

VarieD PictureS Of A RanCH

You don’t need to define them. You just need to spot them as answer choices.

VD - PS - OA - RH

NotebookLM

SYSTEM SHUTDOWN: FINAL REPORT

1. Look for the T (TRouBLe).
2. Check the Shunt (R-L vs L-R).
3. Check the Color (Blue vs Pink).
4. Exceptions: Left Ventricular Hypoplastic Syndrome.
5. Universals: Murmur + Echo.

END OF DIAGNOSTIC CYCLE.

NotebookLM